

Europe's AI Dilemma

The AI Act, Semiconductor Sovereignty, and the Competitiveness Gap with the US

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Executive Summary

Europe has chosen governance over speed. The EU AI Act, which entered into force in August 2024 after more than five years of negotiation, represents the world's first comprehensive statutory framework for artificial intelligence. It establishes tiered obligations across member states and is widely expected to exert a Brussels effect on AI governance globally. Yet the regulatory ambition sits in structural tension with a deepening competitiveness gap: 20 percent of EU enterprises used AI in 2025, but Europe holds less than 3 percent of global AI patents despite producing 18 percent of global AI scientific publications between 2020 and 2024. Private investment remains chronically below US and Chinese levels. European anxieties over dependence on American technology stacks — aired openly at the Munich Security Conference — are mounting. ASML's near-monopoly on extreme ultraviolet lithography equipment gives Europe a singular chokepoint in the global semiconductor chain, but that asset alone cannot substitute for the broader compute infrastructure, frontier model development, and venture capital ecosystem that the continent lacks. [RAG: SCI_TECH_RAG — RAND: The Transatlantic Artificial Intelligence Calculus, 2026-07-12]

1. The EU AI Act: Architecture, Scope, and Implementation Timeline

1. The EU AI Act came into force in August 2024, concluding more than five years of lawmaking and inaugurating a much longer phase of implementation, refinement, and enforcement. [RAG: SCI_TECH_RAG — CSET: The Finalized EU Artificial Intelligence Act: Implications and Insights, 2024-08-01]
2. The Act is the first comprehensive regulation addressing the risks of artificial intelligence through a set of obligations and requirements intended to safeguard health, safety, and fundamental rights of EU citizens, and is expected to have an outsized impact on AI governance worldwide. [RAG: SCI_TECH_RAG — CSET: The EU AI Act: A Primer, 2023-09-26]
3. The regulation works by setting consistent standards for AI systems across EU member states. It covers both domestic and foreign providers placing AI systems on the EU market, creating a de facto global compliance floor for any firm seeking access to the European market. [RAG: SCI_TECH_RAG — CSET: The EU AI Act: A Primer, 2023-09-26]
4. The final legislative text provides for three special transition periods for different categories of rules. For AI models already on the EU market, extended compliance periods apply: general-purpose AI (GPAI) models must comply by 2027; high-risk systems face deadlines extending to 2030. Providers need the intervening period to familiarise themselves with requirements, establish compliance procedures, and bring AI systems into conformity. [RAG: SCI_TECH_RAG — CSET: The Finalized EU Artificial

Intelligence Act: Implications and Insights, 2024-08-01]

5. Carnegie Endowment analysis highlights that for the AI Act to set a global benchmark, the resulting product safety standards must balance detail and legal clarity with flexibility. Autonomous AI agents are increasingly prevalent in cyberspace, and the EU requires a real-time monitoring strategy, investment in AI defenses, and a reduction in its strategic dependence on US frontier models. [RAG: SCI_TECH_RAG — Carnegie: AI and Product Safety Standards Under the EU AI Act, 2024]

6. Regulatory design for frontier systems must grapple with compute thresholds. Today's most compute-intensive models cost tens or hundreds of millions of dollars to train, and targeting regulation at this bracket would automatically exempt smaller developers — a feature considered attractive from a competition policy perspective. Regulatory definitions based on training compute also benefit from being easier to define, measure, and verify than capability-based thresholds. [RAG: SCI_TECH_RAG — CSET: Regulating the AI Frontier: Design Choices and Constraints, 2023-10-26]

2. The Brussels Effect: GDPR as Regulatory Template and the Limits of Normative Power

7. The EU's strategic edge in technology has historically resided in its market, normative, and regulatory powers — what has been described as the "Brussels effect" — illustrated most powerfully by the General Data Protection Regulation (GDPR), though the digital single market remains a work in progress. [RAG: SCI_TECH_RAG — Carnegie: Europe and AI — Leading, Lagging Behind, or Carving Its Own Way, 2020]

8. GDPR entered force on 25 May 2018, requiring digital platforms to obtain active consent when using or sharing user data. The EU regulation has served as a reference model globally; the US has no single equivalent federal data protection law, with the California Consumer Privacy Act as the closest domestic analogue. [CNA[2021-07-27]]

9. The moment GDPR took effect, enforcement actions began immediately. Austrian data privacy activists wasted no time asserting the additional rights the law granted people over the data that companies seek to collect. [CNA[2018-05-25]]

10. European enforcement has continued to produce significant penalties. A Spanish court ordered Meta to pay 479 million euros to Spanish digital media outlets for unfair competition practices and infringement of EU data protection regulation. [CNA[2025-11-20]]

11. The EU governs AI in multifaceted ways while navigating geopolitical, economic, and regulatory concerns. A nuanced understanding of the EU's AI technopolitics is needed, including the ways European efforts to govern AI, foster innovation, and ensure trustworthiness interact with one another. The AI Act regulates the use of AI systems with attention to all three dimensions. [RAG: SCI_TECH_RAG — Carnegie: Charting the Geopolitics and European Governance of Artificial Intelligence, 2024]

12. While the European Commission's digital sovereignty agenda may help advance certain AI developments, analysts caution that regulatory power alone cannot compensate for deficits in compute infrastructure, frontier model development, and private capital. The Brussels effect is a genuine strategic asset, but it is not a substitute for domestic technological capacity. [RAG: SCI_TECH_RAG — Carnegie: Europe and AI — Leading, Lagging Behind, or Carving Its Own Way, 2020]

3. The Competitiveness Gap: Investment, Patents, and the Transatlantic Anxiety

13. The EU produced 18 percent of global AI scientific publications between 2020 and 2024, yet holds less than 3 percent of global AI patents. This research-to-commercialisation pipeline leak has been attributed to structural gaps by the EU itself, the European Investment Bank, and the Draghi report. Twenty percent of all EU enterprises used AI in 2025, but the pipeline from research to commercial scale continues to leak. [RAG: SCI_TECH_RAG — RAND: The Transatlantic Artificial Intelligence Calculus, 2026-07-12]

14. Europe lags behind the United States and China in private investment in AI. In 2016, Europe devoted only between 2.4 and 3.2 billion euros in investment funding, far below US and Chinese levels. Chinese government-backed funding rounds for individual AI companies exceeded Europe's entire annual investment pool. [RAG: SCI_TECH_RAG — Carnegie: Europe and AI — Leading, Lagging Behind, or Carving Its Own Way, 2020]

15. EU member states should acknowledge that the amount of resources required to keep up with the latest AI developments cannot be met by going it alone. There is a clear rationale for a stronger EU-level role and for a more coordinated approach to AI investment, with cross-border networks of excellence and the Horizon 2020 program cited as reference points. [RAG: SCI_TECH_RAG — Carnegie: Europe and AI — Leading, Lagging Behind, or Carving Its Own Way, 2020]

16. European national AI strategies vary considerably. Countries including Czechia, Estonia, Finland, France, Germany, Sweden, and the United Kingdom have each developed national AI strategies. Comparing these approaches and government efforts to shore up Europe's position reveals a patchwork that falls short of the coordinated US and Chinese ecosystem investments. [RAG: SCI_TECH_RAG — Carnegie: Europe and AI — Leading, Lagging Behind, or Carving Its Own Way, 2020]

17. The US and China are ahead in the AI competition. The combination of potentially large economic, societal, and military dividends has propelled countries to join this race and swiftly apply AI across as many sectors as possible. European concerns that the continent is losing ground are well-founded in the investment and patent data. [RAG: SCI_TECH_RAG — Carnegie: Europe and AI — Leading, Lagging Behind, or Carving Its Own Way, 2020]

18. At the Munich Security Conference, discussion revolved around European anxiety over dependence on American firms and the American technology stack. The concern centred on the observation that the US, much like China, has demonstrated a willingness to weaponise aspects of its technological advantage. [RAG: SCI_TECH_RAG — Brookings: Are the US and China Really in an AI Race?, 2022]

19. Historically, a key driver of US cross-border data flow policy has been commercial interests, consumer protection, and criticisms from the EU — particularly over perceived risks associated with EU citizen data transfers to the United States. This tension reflects Europe's structural dependence on US cloud and AI infrastructure. [RAG: SCI_TECH_RAG — Atlantic Council: The US AI Health Data Policy, 2022]

4. Semiconductor Sovereignty: ASML, Supply Chain Dependencies, and the China Variable

20. Europe risks becoming exposed to global tech wars if it does not promote homegrown solutions and address geopolitically risky dependencies in critical technology domains. This risk has been made clear in the case of the global semiconductor value chain, where Europe lacks fabrication capacity commensurate with its design and equipment strengths. [RAG: SCI_TECH_RAG — Carnegie: The EU's Rise as a Defense Technological Power — From Strategic Autonomy to Technological Sovereignty, 2021]

21. In semiconductor manufacturing equipment, ASML is recognised as an exception due to its unique EUV lithography technologies. Alongside Lam Research (deposition and etch), Tokyo Electron (deposition and etch), and KLA (metrology and inspection), ASML occupies a singular position in the global equipment supply chain. [RAG: SCI_TECH_RAG — White House 100-Day Supply Chain Review: Semiconductors, 2021-06-08]

22. Semiconductor production timelines are structurally inflexible: manufacturing a chip can take up to 26 weeks, and sometimes much longer when supply is tight. Production volumes are usually confirmed six months in advance, and it can take months to switch a production line from one type of chip to another. This inelasticity makes European downstream AI compute build-out contingent on long lead times. [RAG: SCI_TECH_RAG — White House 100-Day Supply Chain Review: Semiconductors, 2021-06-08]

23. China has gained market share in six semiconductor equipment segments between 2019 and 2024: CMP tools, etch and clean tools, deposition tools, handlers and probers, test tools, and advanced packaging tools. The United States gained share in only two segments — advanced packaging and other. This pattern of Chinese advancement compresses the window in which European equipment dominance translates to geopolitical leverage. [RAG: SCI_TECH_RAG — CSET: Inside Beijing's Chipmaking Offensive, 2025-07-14]

24. Despite China's equipment gains, Chinese firms continue to struggle with advanced semiconductor lithography systems — the technology most crucial to self-sufficiency at the cutting edge. This is the domain where ASML's EUV monopoly remains intact. [RAG: SCI_TECH_RAG — CSET: China Lags in Chip Lithography, 2025-07-14]

25. The Chip Security Act in the United States reflects the difficulty of translating hardware chokepoints into durable policy leverage. Earlier proposals to mandate kill switches in chips faced strong industry resistance due to concerns over cost, reliability, and the potential effect on international sales. These ideas either stalled or were watered down beyond usefulness. [RAG: SCI_TECH_RAG — Atlantic Council: How the Chip Security Act Could Usher in an Era of Trusted Trade, 2022]

26. International allies are encouraged to adopt coordinated approaches to disclosing vulnerabilities in compute infrastructure. Encouraging transparency from cloud providers can shift the incentive structure of the AI compute industry — an area where European cloud sovereignty initiatives have direct relevance. [RAG: SCI_TECH_RAG — Atlantic Council: Securing Cloud Infrastructure AI, 2022]

Outlook

Europe enters the second half of the 2020s with a paradox at the heart of its AI strategy: the world's most sophisticated AI regulatory architecture exists alongside one of the sharpest competitiveness deficits among major economic blocs. The Brussels effect gives the EU real normative leverage — the AI Act's extended compliance timelines through 2027-2030 will force global firms to build compliance infrastructure for the European market, and GDPR enforcement actions demonstrate that the penalties are real. ASML's EUV monopoly remains Europe's single most consequential technology chokepoint in the global AI stack.

Yet the Draghi report diagnosis — that the research-to-patent pipeline leaks, that private AI investment is structurally below US and Chinese levels, and that no individual member state can match the scale of resources required — remains unresolved. The anxiety expressed at the Munich Security Conference over dependence on the American technology stack reflects a genuine structural condition. China's advancing market share in semiconductor equipment segments narrows the window in which European equipment dominance can be translated into broader strategic leverage.

The critical variable for the decade ahead is whether the EU can translate its regulatory architecture into a genuine innovation engine — protecting its research base, encouraging governments to be early adopters, fostering the startup ecosystem, and expanding compute infrastructure — or whether the Brussels effect remains a normative export while the actual frontier of AI development continues to widen between Europe and the US-China dyad.

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